

Sohrab Pirhadi

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RESEARCH PROFILE

Motivated researcher with a background in Machine Learning, Deep Neural Networks, and Natural Language Processing. Skilled in ensemble methods and feature engineering for complex, interdisciplinary datasets. Research interests focus on developing explainable and interpretable AI models bridging NLP, chemometrics, and real-world scientific challenges.

RESEARCH INTERESTS

- Machine Learning
- Deep Neural Networks
- Explainable & Interpretable AI
- Natural Language Processing
- Information Extraction
- AI in Health/Chemometrics

EDUCATION

- Master of Computer Science - Artificial Intelligence** Sep 2019- Jun 2022
Institute for Advanced Studies in Basic Sciences (IASBS) Zanjan, Iran
Thesis: "Sentence-Level Relation Extraction Using Deep Neural Networks". [DOI: 10.13140/RG.2.2.31721.30568](https://doi.org/10.13140/RG.2.2.31721.30568)
Supervisors: Dr. Ebrahim Ansari
Related Courses: Advanced Machine Learning, Data Mining, Deep Learning, Advanced Algorithm Design
GPA: 17.15/20
- Bachelor of Software Engineering - Information Technology** Sep 2013 – May 2018
Institute for Advanced Studies in Basic Sciences (IASBS) Zanjan, Iran
Thesis: "Dimension Reduction in Data Mining"
Supervisor: Dr. Zahra Narimani

WORK EXPERIENCE

- ML Engineer | Gahar Research Group** Jun 2026 – Now
National University of Borujerd (ABRU) Borujerd, Iran
Developing a bilingual RAG chatbot for grounded, document-based question answering.
- Research Assistant | Deputy of Research** Jun 2024 – Jun 2026
National University of Borujerd (ABRU) Borujerd, Iran
Collaborated on multiple interdisciplinary research projects in partnership with the University of Tehran.
- Research Assistant | Artificial Intelligence Lab** Oct 2022 - Jul 2023
Institute for Advanced Studies in Basic Sciences (IASBS) Zanjan, Iran
Resulted in a publication stemming from thesis research.
- Teaching Assistant | Dr. Mahdi Vasighi** Mar 2021 - Jul 2021
Institute for Advanced Studies in Basic Sciences (IASBS) Zanjan, Iran
Assisted in teaching the Computational Data Mining course.

PUBLICATIONS

Peer-Reviewed Journal Articles

- Sohrab Pirhadi**, N. Shariatifar, M. Pirhadi. "Identification of Animal Species in Meat Products Using Chemometrics-Based High-Performance Liquid Chromatography: A Systematic Review." *Journal of Biochemicals and Phytomedicine*, 2024. [DOI: 10.34172/jbp.2024.11](https://doi.org/10.34172/jbp.2024.11)
- M. Pirhadi, N. Shariatifar, **Sohrab Pirhadi**. "Classification of Meat Species in Sausage Products Using FTIR Spectroscopy and Multivariate Analysis for Food Authentication." *Results in Chemistry*, 2025. [DOI: 10.1016/j.rechem.2025.102871](https://doi.org/10.1016/j.rechem.2025.102871)
- M. Pirhadi, N. Shariatifar, **Sohrab Pirhadi**, S.M. Khodaei, Y. Mazaheri. "Developing Infrared Spectroscopy Methods for Identification of Food Fraud and Authenticity: A Review." *Journal of Biochemicals and Phytomedicine*, 2024. [DOI: 10.34172/jbp.2024.12](https://doi.org/10.34172/jbp.2024.12)

Conference Proceedings

- **Sohrab Pirhadi**, B. Niroomand, E. Ansari. "Finding the Sweet Spot: An Empirical Study on Dataset Size, Performance, and Efficiency in Relation Extraction." *29th International Computer Conference (CSICC)*, IEEE, 2025. [DOI: 10.1109/CSICC65765.2025.10967456](https://doi.org/10.1109/CSICC65765.2025.10967456)

Under Review / In Preparation

- **Sohrab Pirhadi**, B. Niroomand, et al. "Meat Species Authentication in Processed Sausages using SMOTE-Augmented HPLC/MS Peptide Fingerprints: An Explainable Machine Learning Approach with Feature Selection and Biomarker Interpretation." *Food and Bioprocess Technology — Under Review*, 2026.
- **Sohrab Pirhadi**, E. Jafari, et al. "Estimating Longitudinal Dispersion in Natural Rivers Through Explainable Machine Learning and Hydraulic Feature Engineering". *Journal of Contaminant Hydrology — Submitted*.
- E. Jafari, et al., **Sohrab Pirhadi**. "Boosting the Prediction Performance of Longitudinal Dispersion Coefficient for Pollutant Transport in Natural Rivers Using Machine Learning Algorithms." *Journal of Contaminant Hydrology — Submitted*.
- B. Niroomand, **Sohrab Pirhadi**, M. Shirazi. "A Comparative Analysis of Ensemble Learning Methods for Classification Tasks on Benchmark Datasets"—*Manuscript in Preparation*. Preprint: [DOI: 10.13140/RG.2.2.30835.64809](https://doi.org/10.13140/RG.2.2.30835.64809)
- B. Niroomand, **Sohrab Pirhadi**, M. Shirazi. "Random Forest on GPU: A Comparative Survey with Empirical Reconstruction and Benchmarking"—*Manuscript in Preparation*.

LANGUAGES

- English: TOEFL iBT planned (Dec 2026)

CERTIFICATIONS

- Data Science in R – tutored by Dr. Mohammad Shokouhi Yekta (Apple Inc., Stanford University). Dec 2019
- [Exploratory Data Analysis for Machine Learning](#) - IBM Dec 2023
- Reviewer for manuscripts in a peer-review journal, (*BEEI*), ISSN: 2089-3191, e-ISSN: 2302-9285 Mar 2025
- Certificate of Presentation – 29th CSICC, Sharif University of Technology, 2025. Dec 2025

SKILLS

Technical Skills

- **Programming:** Python, C/C++, JavaScript
- **Machine Learning Frameworks:** PyTorch, TensorFlow, Keras, LangChain, Hugging Face, Transformers
- **Data Science & Visualization:** Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn
- **Database Management Systems:** SQL (PostgreSQL, MySQL), NoSQL (MongoDB, Redis)
- **DevOps & Version Control:** Docker, Git, GitHub

Analytical & Research Skills

Problem solving and Algorithm design, Data preprocessing and feature engineering, Model evaluation and performance optimization, Literature review and comparative analysis, Scientific writing and presentation, Academic writing and peer collaboration.

REFERENCES

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| <ul style="list-style-type: none">• Dr. Ebrahim Ansari (Thesis Supervisor)
Assistant Professor, Department of Computer Science
Institute for Advanced Studies in Basic Sciences
Email: ansari@iasbs.ac.ir | <ul style="list-style-type: none">• Dr. Parvin Razzaghi
Assistant Professor, Department of Computer Science
Institute for Advanced Studies in Basic Sciences
Email: p.razzaghi@iasbs.ac.ir |
| <ul style="list-style-type: none">• Dr. Mahdi Vasighi
Assistant Professor, Department of Computer Science
Institute for Advanced Studies in Basic Sciences
Email: vasighi@iasbs.ac.ir | <ul style="list-style-type: none">• Dr. Zahra Narimani
Assistant Professor, Department of Computer Science
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